

- They are evidence of extensions required to serve clinical purposes in identified situations.
- It is preferable to minimise incompatibilities with these classifications.
- Take account of classifications other than ICD that are in wide clinical use for conditions in scope for this chapter.
- Take account of advice, solicited and proffered.
 - Increased attention to distinctions pertinent to treatment choices and to outcomes, including disability.

These include allowing identification of clinically and prognostically important aspects of fractures (notably whether they extend into a joint) and organ/vessel injuries (degree). Some conditions are much more important when bilateral, and in such instances side has been proposed as precoordinated entities (e.g. injury of the eyes). The United States' clinical modification of ICD-10 (ICD-10-CM) was particularly valuable in this regard, as its injury chapter makes many distinctions, beyond ICD-10, which follow or are consistent with credible and widely used clinical classifications relevant to injury treatment and outcome.

Increased attention has been given to injury conditions specific to childhood (e.g. greenstick and epiphyseal fractures) and to injury conditions that are indicative of possible intentional injury (e.g. posterior rib fractures, 'bucket-handle' and 'corner' fractures).

The work was done with the awareness that this chapter is not primarily used to code the Underlying Cause of Death.

The morbidity use case is particularly important for this chapter.

3.2.23 Chapter 23 – External causes of morbidity or mortality

3.2.23.1 Chapter 23 – Structure of Chapter 23

The general hierarchy of Chapter 23 consists of the following sections:

- 1st level - Intent of external cause (unintentional, intentional self-harm, assault, undetermined intent and intent pending.)
- 2nd level - Broad category of the mechanism of the external cause
- 3rd level - More specific mechanism and objects/substances involved in causing injury
- 4th level - Further characterisation of the external cause

Other sections include Exposure to extreme forces, Maltreatment, Legal intervention, Armed conflict and Causes of health care related harm.

3.2.23.2 Chapter 23 – Rationale for Chapter 23

The main aim of the changes was to provide a more uniform coding structure while still maintaining high compatibility with ICD-10. The changes to the traffic injury categories are aimed at simplifying code selection, while the section on Operations of war and armed conflicts has been revised to capture the more current situations of armed conflicts. Another enhancement has been to produce a single, hierarchical list of noxious substances to serve the Injury and External Causes chapters.

All mechanisms/objects codable for all intents:

- More uniform code structure
- Revised 'Intent' dimension (N.B. Intent pending; ISH: suicidal/non-suicidal)
- Retain transport codes, but expand vehicle types
- Expanded Place of Occurrence codes
- Expanded and revised Activity dimension (N.B. work-relatedness)
- Revision of Complications of Medical & Surgical Care
- Expanded Legal/War Codes
- Improved provision for maltreatment syndromes
- Introduction of additional dimensions (optional)
- Revision of External Cause index, rules and guidelines
- Provide for Mortality, Morbidity, Lower Resource Settings, Research

Progress has been made on all of these points, though constrained in some respects, particularly for the mortality use-case (due to the tight constraints on code-space combined with the lack of provision for postcoordination/cluster-coding). A section on limitations is at the end of these notes.

Notes provided here focus on several of these points; additional material will be provided on other aspects on request. Comments are also provided here on the two main issues that involve both the External Causes chapter and the Injury chapter (both also involve Chapter X 'Extension codes'): substances; complications of care (Safety & Quality).

Transport

Four dimensions are implicit in the ICD–10 range V01-V89: injured person's mode of transport (e.g. motorcycle), whether the injurious event occurred in road traffic (if so, the resulting injury is a road injury), the injured person's role (e.g. passenger), and what other type of vehicle was involved, if any (counterpart). All four dimensions are required for a revised structure that is conceptually equivalent to the ICD–10 'transport accidents' module at four- character level.

All four dimensions have been precoordinated in the Unintentional transport injury module. This produces a structure with high back-compatibility with ICD–10 at four-character level. It preserves all top-level modes of transport categories (some now split) and the four conceptual dimensions (mode; and for land transport modes: whether in traffic, transport user role and counterpart).

In recognition of code-space limitations, and of the fact that most transport injury cases are unintentional, precoordination of transport cases in the other main intent blocks (intentional self-harm, assault, undetermined intent, and intent pending) is limited to intent by mode of transportation. However, the other dimensions are available for optional use.

The revised transport block includes changes made to resolve problems identified with the ICD– 10 transport section.

- Split several modes of transport to enable identification of important and emerging types that cannot be identified in ICD–10.
- Refined and revised terms and definitions (for clarity, to fill gaps in the set provided in ICD–10 and to improve comparability with terms used internationally for road safety).
- Various other revisions (e.g. of types of vessel in water transport section). Note that the coordination order has been altered from the equivalent in ICD–10, from: mode, counterpart, then user role and traffic status combined to: mode, traffic status, user role, counterpart.

The main reason for this change was to simplify the selection of ‘traffic accident’ categories, which are frequently required when reporting road injury.

War and armed conflict

A revised classification is provided for inclusion as the expansion of intent category Armed conflict (Operations of war in ICD–10). The classification largely follows the expansion of Y36 in the United States’ clinical modification of ICD–10 (ICD–10-CM). This follows the four-character categories in ICD–10 and provides subdivisions, which follow inclusion notes given in ICD–10. In addition, sub-categories are provided to distinguish whether the injured person was military or civilian.

The rubric has been altered by the addition of ‘...and armed conflict’ to ‘Operations of war’, and the inclusion term has been altered accordingly. ‘War’ and ‘civil insurrection’ (which also formed part of the inclusion term) were not defined in ICD–10. The use of a term broader than ‘war’ is considered desirable because war, in the sense of formally declared armed conflicts between nation states (or subnational entities) has become uncommon. Armed conflicts of a range of types and intensities, while tending to become less common, remain much more numerous than wars. Restriction of use of this category to declared wars, and/or to armed conflicts that meet a commonly used criterion of intensity (1,000 or more battle-related deaths in a calendar year = war) was thought to be unduly restrictive. The alternative proposed here is to also include injuries due to ‘Minor’ armed conflicts, defined as those resulting in 25 to < 1,000 battle-related deaths in a calendar year. Application of the definition is aided by the existence of a publicly accessible database listing conflicts found to satisfy it.

Crossover issues

These are matters that affect both the injury and the external causes chapters, and other parts of the ICD.

Toxic effects of substances

Toxic effects of noxious substances appear in ICD–10 at several points, in the Injury and External Causes chapters, and in other chapters. Code lists at those points differ in specificity and are not completely consistent. A design aim for ICD–11 is to produce a single, hierarchical list of noxious substances to serve all of the purposes required for the Injury and External Causes chapters. The benefits of this are: external source(s) can define-by-example the inclusions of the ICD–11 list; and if the external source(s) are actively updated, then this provides a way for the ICD–11 coverage of substances to remain current.

The term 'Harmful effects' is used for all types of harm resulting from harmful chemical effects of substances of all types. It is recognised that other terms, such as 'toxic effect', 'poisoning', 'chemical corrosion' and 'envenomation' are sometimes used in the context of particular substances. These terms will be included as synonyms and subordinate terms where in common use. A number of sources were consulted, including ICECI Objects & Substances dimension; Anatomical Therapeutic Chemical (ATC) classification; TAG-IEG advisory groups on drugs and poisons; Quality and safety TAG; SNOMED; IPCS INTOX.

The list has two main hierarchical levels.

The first, with 16 categories, is conceptually related to the code-list that is present in ICD–10 at X40-X49 (Accidental poisoning by and exposure to noxious substances) and the equivalent points in the Intentional Self-harm and Undetermined intent code-blocks. The list results from application of these principles:

- It should have only a few categories. This is necessary for practicability, especially in the context of cause of death coding and because the block structure of the external causes chapter has the effect that each additional category adds several rows.
- The categories should refer to substances or classes of substances that are important causes of mortality or morbidity.
- As many as possible of the categories should be sufficiently specific so meaningful as reporting groups. (By comparison, several categories in the ICD–10 blocks such as X40-X49 are so broad as to be difficult to interpret).
- The several main contexts of exposure were kept in mind when specifying categories (i.e. recreational/street use; clinical use; self-harm; industrial and other exposures).

The 16 categories, either alone or combined with others, allow backwards comparability with eight of the ten categories in ICD–10 X40-X49 (and the equivalent groups in the ISH and Undetermined intent blocks). The only exceptions are two residual groups: '...other gases and vapours' and '...other and unspecified chemicals and noxious substances'. The second level provides categories (n=381), with about the same number and specificity of substances that are provided for in the injury and external causes chapters of ICD–10. It includes all of the categories of substances that are specified in the 'Cause of harm' component of the Quality and Safety TAG classification.

Some categories have been added: to allow for pharmacological innovation and changes in drug use (e.g. synthetic cannabinoids); to reflect additions to ICD–10 made in its clinical modifications (e.g. more specificity concerning anticoagulants); to allow more specific identification of prominent drugs (e.g. paracetamol); to provide for additional widely-used recreational drugs (e.g. Cathinone, the main active agent in khat); and on advice from other TAGs (e.g. types of substance added by the Safety and Quality TAG). We anticipate that more categories will be added in future updates, to reflect changes in drug availability and use.

A more comprehensive list of substances (a superset of the hierarchical list), with synonyms for many of the entries, will be provided in Chapter X 'Extension codes'. That list shares the same hierarchical structure as the precoordinated codes. It also takes account of the ICD–11 Supplementary Classification of Contact Allergens prepared by the Dermatology TAG.

Entries in the Extension codes substances list will be specified in terms of equivalent terms in SNOMED-CT.

Complications of care (Quality and Safety)

This section briefly describes the model for coding complications of care that has been developed by the Quality and Safety Topic Advisory Group (TAG).

The model has three parts, each of which must be coded. The postcoordinated codes for all the parts must be designated as belonging to a cluster. The three concepts are:

1. The resultant injury or harm;
2. The cause of harm; and
3. The 'Mode/Mechanism' of harm. Classifications and code-sets have been developed for (2) and (3) by the Quality and safety TAG. The categories have been entered into the External Causes chapter. The resultant harm (1) is to be coded by using the most appropriate disease or injury code from any part of ICD–11.

The construct would, in principle, fit well into ICD–11 as follows:

1. Resultant injury or harm. Code selected from anywhere in ICD–11.
2. The cause or 'Mode' of harm: Code selected from the relevant block in External Causes chapter
3. 'Mode/Mechanism' of harm

Sanctioning rules lead coders to the subset of 'Mode' codes that are relevant, given the selected 'Cause' (e.g. if 'Cause' is a drug, then the relevant 'Modes' are categories such as overdose and underdose).

3.2.24 Chapter 24 – Factors influencing health status or contact with health services

This chapter should not be used for international comparison or for primary mortality coding.

3.2.24.1 Chapter 24 – Structure of Chapter 24

This chapter has two major sections: - Reasons for contact with the health service - Factors influencing health status

The general hierarchy of Chapter 24 consists of the following axis:

- 1st level - Broad category of a particular health status or service
- 2nd level - Specific condition

3.2.24.2 Chapter 24 – Rationale for Chapter 24

Initially, the Functioning Topic Advisory Group for ICD–11 (fTAG) was tasked with the review of the Factors Chapter. They were to evaluate the necessity of each of the 801 codes and propose a revised hierarchical structure for the essential content that would remain. This content was to be both clinically relevant and use-friendly as well as allowing the necessary space for expansion using the extension codes, as necessary. fTAG organised a review that identified the major 'types' of codes as 'diagnostic', 'interventional', 'contextual factors' and

‘other/debatable’. This review was combined with the general structure of the ICPC2 classification section on ‘social problems’ and a new organisation was designed that combined the ICPC2 hierarchy with the ICD-11 codes. For the ICD-11 MMS, a shoreline exercise was then undertaken on the new structure to decrease granularity seen as unnecessary.

3.2.25 Chapter 25 – Codes for special purposes

3.2.25.1 Chapter 25 – Structure of Chapter 25

This chapter consists of two blocks:

- International provisional assignment of new diseases of uncertain aetiology, containing the international emergency codes
- National provisional assignment of new diseases of uncertain aetiology, containing codes for use by individual countries

3.2.26 Chapter 26 - Supplementary Chapter Traditional Medicine Conditions - Module 1

‘Traditional Medicine Module 1’ (TM1) chapter is a new supplementary chapter for optional use in ICD, and as such is referred to as the ‘TM1 chapter’. The rationale for its inclusion in ICD-11 is to enable Traditional Medicine health services and encounters to count and be counted nationally and internationally. The Module in this chapter in its current form refers to disorders and patterns which originated in ancient Chinese Medicine and developed throughout history to incorporate contemporary science and technology. These disorders and patterns are commonly used in China, Japan, Korea, United States of America, Australia, Europe and elsewhere around the world. The classification rubrics represent a unified set of harmonised Traditional Medicine disorders and patterns from national classifications from China, Japan and Korea. Future Modules may be developed for other forms of Traditional Medicine practices.

Scope:

This chapter has currently been designed for morbidity recording and reporting. It must not be used for mortality coding and reporting.

Content and structure:

The content and structure of the TM1 Chapter represent a common language developed jointly through the international cooperation of Traditional Medicine clinicians, researchers, academics and classification experts to enable international comparability of practice and reporting of morbidity in Traditional Medicine. Standardisation of this TM1 classification will allow clinical documentation in different countries to incorporate the same concepts and enable coders and users to extract comparable morbidity data from that documentation. Coders must also be guided by rules which reflect the clinical diagnostic decision-making process. However, the rules are relatively flexible to allow for national adaptations and research questions concerning relationships between diseases, disorders and patterns to be framed from a number of different angles.

The English terms do not necessarily represent the most common translation of the TM terms in Chinese, Korean or Japanese. Where the best fit English TM translation resulted in the same term as used in Western Medicine, it was necessary to indicate a difference

between the Western Medicine (WM) concept and TM concept where the same term had different definitions in TM and WM. This difference in definition is indicated by the use of (TM1) for disorders and patterns throughout the TM chapter.

Terminology:

The Supplementary Chapter Traditional Medicine Conditions, Module 1, uses the terms disorder and pattern to describe concepts. This is different from the concept descriptions in the Western Medicine chapters which refer to diseases (clinical pictures) and syndromes (clinical presentations). The TM1 chapter is divided into separate sections for disorder and pattern to emphasise the independence of these concepts.

Definitions

A disorder in traditional medicine (disorder (TM1)) refers to a set of dysfunctions in any body system which is judged from associated signs, symptoms or findings. Each disorder (TM1) may be defined by its symptomatology, aetiological explanation based on traditional medicine, course and outcome, treatment response or linkage to interacting environmental factors. A disorder (TM1) is a clinical picture that is relatively stable and reflects the local pathology and related specific manifestations commonly found in the anatomy and function of the affected individuals.

A pattern in traditional medicine (pattern (TM1)) refers to the manifestation of the patient's health condition at a given moment in time including all findings which may include:

- *Symptomatology*: pattern of specific and non-specific signs, symptoms or unique findings by traditional medicine diagnostic methods, including the taking of the pulse, examination of the tongue, abdominal examination and other methods that reflect the systemic response of the patient in a dysfunctional condition.
- *Constitution*: the characteristics of an individual, including structural and functional characteristics, temperament, ability to adapt to environmental changes, or susceptibility to various health conditions.

A pattern (TM1) is a clinical picture that is relatively temporary, reflects on the systemic response of the patient and combined pattern of specific and non-specific manifestations that usually hold a multifactorial relationship with the local pathology and the constitutional traits of the patient. A pattern may show individual difference even in the individuals affected by the same pathology that may be further analysed by the theoretical frame of Traditional Medicine.

Traditional Medicine disorder and pattern are named after the body structures, causal explanations, properties and severity which present for clinical investigation and diagnosis. TM1 pattern may denote an individually different pattern (TM1) of systemic responses to the WM disease or TM1 disorder. Pattern is a concept unique to TM1 and may be different from TM1 disorder in the following ways:

Table 1: Characteristics of Traditional Medicine Disorders and Patterns

Distinguishing Feature	Disorder in Traditional Medicine	Pattern in Traditional Medicine
Constant/ Temporary	A clinical picture that is relatively constant throughout the duration of that disorder	A clinical picture that is relatively temporary
Constant Pathology/ Temporary Response	Usually delivers information reflecting the constant pathology	Usually delivers information reflecting the temporary overall manifestation or response of the patient
Specific/ Non-specific	A concept that summarises findings that are specific to the pathologic process under investigation	The combination of the manifestations that encompasses both specific symptoms/signs and non-specific findings
Linear/ Multifactorial	May be applied for a time span. A disorder coding may be based on the main pathologic process which may show a causal relationship with the main manifestations in the patient	A pattern may be applied for a specific time span, too. However, a pattern code is based on the summarised whole picture that may be observed in the patient based on the perspectives of traditional medicine theories. A pattern is recognized based on the analysis of the systemic findings in the patient's body and mind which reflect the pathologic processes, responses to the pathologic processes, other concomitant findings, and innate or acquired constitutional traits of the patient
Commonality/ Individuality	Used to describe the general characteristics considered to be relatively common to the population suffering from one particular disorder	Used to describe the individual characteristics considered to be relatively specific to the patient at that time
General/ Theoretical	Usually described with general terms of anatomy and physiology together with terms of signs and symptoms	Usually described with terms of the traditional medicine theories that are used to summarise the underlying mechanism in the patient such as yin and yang balance, cold and heat, meridian, or constitution

3.2.27 Section V – Supplementary section for functioning assessment

This section is new. The list of 47 entities in this section is intended for assessment and scoring in the context of ICD. It is using ICF functioning domains of high explanatory power (ICF Annex 9). The categories are intended to be used as a set. The set has been defined in a way that general and domain specific summary scores can be calculated using the WHO

Disability Assessment Schedule 2.0 (WHO DAS 2.0) or the WHO Model Disability Survey (MDS).

3.2.28 Chapter X - Extension Codes

This chapter is new. Extension codes are envisaged as providing the basis for postcoordination of ICD–11 codes, being the repository for all codes in a linearisation that are not eligible for use as stem codes.

The different lists provide additional codes for clinical use as well as for injury research, device safety, drug safety, patient safety and cancer registration.

These codes are for optional use and will be used more likely in morbidity than in mortality.

3.3 Multiple Parenting

An entity may be correctly classified in two different places, e.g. by site or by aetiology. For a disease like oesophageal cancer this would mean that it could be classified to cancers (malignant neoplasms) or to conditions of the digestive system. In the same way, cerebral ischaemic conditions could be classified to the vascular system – where the problem arises – or to the nervous system – where the ischaemia impacts and manifests with symptoms.

Indications of multiple parenting:

- ‘Excludes’ or ‘Code elsewhere’ note
- Display of multiple parents in Foundation Component view
- Display of multiple parents in tabular list. Example for oesophageal cancer: primary parent malignant neoplasm will appear in black and the digestive system for the oesophageal cancer in grey

In the Foundation component, ‘excludes’ notes for these examples will mention possible parents (multiple parents). However, for the tabulation of statistical outputs from any tabular list, there can be only one parent for primary tabulation. When there are such multiple parents, in the Foundation Component view both parents will be displayed the same way. However, in a tabular list, the primary parent place will show the entity and its parents in black, and the secondary possible parent in grey.

Every time an entity is parented elsewhere, it will continue to show the code from the primary parent. The primary parent is sometimes referred to as the ‘Tabular list parent’.

3.4 The Content Model

The Content model is a structured framework that defines each entity found in the ICD in a standard way. The purpose of the Content model is to present the background knowledge that provides the basis for the description of each ICD entity in a systematic way to allow for computerisation. ICD–11 holds all of its content in the Foundation Component. Here, every entity is specified by a description, machine readable properties that have values, and one or more parent-child relationship(s). Additional links provide information for postcoordination. All of this multi-dimensional information is then combined to form a list with mutually exclusive categories - the tabular lists. The Foundation Component includes

information on where and how a certain entity is represented in a tabular list. An entity might become a grouping, a category, or just a term that is, for example, listed in the index.

Each ICD entity can be seen from different dimensions. The Content Model represents each one of these dimensions as a 'property'.

The key components of the descriptions of disease are included as different properties within the Content model. The main properties of the Content model are:

1. ICD Concept Title
2. Hierarchy, Type and Use
 - Parent
 - Type
 - Use
3. Textual Definition(s)
 - Description (short)
 - Definition (long)
4. Terms
 - Index terms
 - Synonyms
 - Inclusion terms
 - Exclusion Terms
5. Clinical Descriptions
 - Body System(s)
 - Body Part(s) (Anatomical site(s))
 - Manifestation Properties
 - Signs and Symptoms
 - Findings
 - Causal Properties
 - Aetiology Type
 - Infection (agents)
 - Injury (mechanisms)
 - Risk Factors
 - Genomic Characteristics
 - Temporal Properties
 - Severity Properties
 - Functional Properties
 - Specific Condition Properties
 - Treatment Properties
 - Diagnostic Criteria

For each ICD entity, various properties can be given if necessary to reach the correct coding result. A coder can use the properties to better understand a condition and how to code it, and the properties allow a coder to determine where a condition is placed. Only properties that are absolutely necessary have been defined in order to avoid the necessity of frequent updates. Additions of property values at the international level are managed through the

regular update cycle whenever coding problems indicate the necessity to do so. The following is an example of the core properties:

ICD entity: Invasive ductal carcinoma of breast

Properties	Value
Anatomy	Breast
Morphology	Invasive ductal carcinoma

The full range of different values for each given property is predefined using standard terminologies and ontologies. This range of values is called a 'Value set'.

Descriptions of ICD–11 entities inform analysts and translators about the meaning of an entity and its descriptive characteristics. There are two different types of descriptions: a short description (maximum of 100 words) that is available in both the content model and the tabular list, and a detailed description with comprehensive detail at the level required for each entity. The detailed description is labelled 'additional information' and appears only in online electronic versions. Coders are cautioned not to use the descriptions for coding. Coders must assign codes based on the diagnosis(es) documented by a clinician. However, the descriptor information that is included for the individual entities of the ICD-11 provides users of the ICD with clear insight regarding the intended meaning and scope of rubrics or terms in the tabular list and the Foundation Component. The descriptors guide translators, coders, and users of coded data. The goal is to enhance the comparability, consistency, and interpretation of coded information for everyone, everywhere. There are four levels of descriptor information in the ICD–11 content model:

- **Fully Specified Term** - This is an unambiguous title that does not assume context. For example, 'systemic illness with predominant gastrointestinal diarrhoeal symptoms attributable to vibrio cholera', as opposed to 'cholera' or 'other' (where the meaning of other would have been clear from the hierarchical context).
- **Short Description** - The short description is a characterisation (maximum of 100 words) of the entity that states things that are always true about a disease or condition and which are necessary to understand the scope of the rubric. Descriptions do not necessarily contain elements intended for use in common epidemiology or things that are clinical criteria.
- **Additional Information** - This is a text field that is not mandatory, but that may contain additional information about, or characteristics of the diseases or conditions included in the entity. This text field provides more context for the entity. For example, the most common epidemiologic circumstances, putative or highly suspected aetiologic agents, or other information that may not always be true but may be common, typical, or expected.
- **Clinical or Diagnostic Criteria** - It is intended to contain one or more scenarios of clinical criteria and characteristics that would be sufficient to diagnose the condition(s) or syndrome(s) of the given class or concept.

Such scenarios will contain multiple variations, or embedded logic to accommodate 'x out of n' variations, that are necessary or useful to make the diagnosis. For example, a myocardial infarction (MI) in high-resource diagnostic settings would typically include a longitudinal pattern of cardiac enzymes, specific electrocardiogram changes, and stereotypical

symptoms. However, only two out of these three needs to be present as there are such things as ‘silent MIs’ (without symptoms) and similar variations.

It is expected that these scenarios will be divided over technology capabilities. For example, diagnosing a myocardial infarction in the high-resource diagnostic settings would likely involve different technology and criteria than in low-resource settings. ICD diagnostic criteria draw on various WHO guidelines that have identified diagnostic rules. Extensions to the ICD, as specialty linearisations, may use such diagnostic rules.

The ICD-11 architecture, and the future evolution of this component of information, could eventually serve decision algorithms based on these criteria. Assignment of diagnoses and conditions could automatically be proposed from data arising in electronic medical records.

Diagnostic criteria describes diagnostic methodology that determines how health providers diagnose cases that are classified to an entity. It contains the core diagnostic information necessary and sufficient to describe a category, and enables the digital representation of the diagnostic algorithms using standardised terminology and other elements as appropriate. There may be different sets of diagnostic criteria for different settings. Diagnostic criteria will draw on the content of other attributes.

3.5 Language independent ICD entities

ICD-11 entities are language independent. All entities have a unique identifier or uniform resource identifier (URI) and have a specific place in a hierarchy of groups, categories and narrower terms. The maintenance of the ICD-11 on an international level is handled in the English language but the content model of the ICD-11 is language independent and allows binding of any desired language to the elements of its foundation. In this way, an international translation base facilitates translations and multilingual browsing. The URI remains valid independent of whether an ICD entity is still valid or has been retired. The hierarchical structure of groups, categories, subcategories, and inclusions (parents, children and narrower terms) serves also as a language independent backbone for translations of ICD. This structure is essential when building an index in a local language. It helps (in conjunction with the ICD translation platform) to identify the things that need to be translated in order to be able to address ICD categories with terms reported in the local language.

3.6 Innovation to mortality coding in ICD-11

Mortality coding instructions of ICD have matured over time and basically has been maintained for ICD-11, while the text has been formatted using easier wordings to enhance common understandings and standardised implementation. Major changes in the classification have been incorporated in the mortality coding instructions.

New concepts or terminology of ICD-11, for instance postcoordination or cluster coding, ‘code also’ and ‘use additional code if desired’ instructions will function to capture further information reported on the death certificate. In ICD-10 mortality coding, multiple cause coding, several modification rules in Step M4, or certain flags used in automated coding systems has been used to capture details reported on the death certificate and to facilitate accurate selection of underlying cause. And in this sense, it is considered that the function of postcoordination etc. has been embedded in ICD-10 mortality coding practice, while in

ICD-11 the concepts are more evident and several new instruction were added to inform on how to apply these new concepts (see Part 2 of this Reference Guide).

The following table used in Step M1 for coding complications of diabetes mellitus is provided for optional use. This list is not a complete list of possible complications of diabetes mellitus, and is intended not to be updated but kept for users who are interested in consistency between ICD-10 and ICD-11 coding.

TUC diabetic complication in ICD-11	If desired, postcoordination may be used to specify complication of diabetes mellitus with mention of:
(ketoacidosis)	5C73 Acidosis 5C50.G Trimethylaminuria MA18.Y Other specified abnormal findings of blood chemistry
(renal complications)	GB40-GB4Z Glomerular diseases GB61 Chronic kidney disease GB6Z Kidney failure, unspecified MF54.0 Smooth contracted kidney GB90.4Z Renal tubular function disorders, unspecified
(ophthalmic complications)	9A96.Z Anterior uveitis, unspecified 9B10.Z Cataract, unspecified 9B65.Z Posterior uveitis, unspecified 9B78.1 Background retinopathy and retinal vascular changes 9B78.2 Other proliferative retinopathy 9B7Z Disorders of the retina, unspecified
(neurological complications)	8C12 Certain specified mononeuropathies 8C1Z Mononeuropathy of unspecified site 8C0Z Polyneuropathy, unspecified 8C4Y Other specified disorders of nerve root, plexus or peripheral nerves 8C7Y Other specified primary disorders of muscles 8D8Z Disorders of autonomic nervous system, unspecified
(peripheral circulatory complications)	BD40.0 Lower limb atherosclerosis BD4Z Chronic arterial occlusive disease, unspecified EE80.1 Necrobiosis lipoidica MC85 Gangrene
(other specified complications)	ME60.2 Ulcer of skin of uncertain nature, specified as lower limb FA2Z Inflammatory arthropathies, unspecified MG30.5Z Chronic neuropathic pain, unspecified
(coma)	when reported as the cause of: MB20.1 Coma
(ophthalmic complications)	9C81.Z Palsy of unspecified ocular motor nerve

**TUC diabetic complication
in ICD-11**

If desired, postcoordination may be used to specify
complication of diabetes mellitus

(neurological
complications)

[9D90](#) *Vision impairment including blindness*

[8E7Y](#) *Other specified diseases of the nervous system*

(peripheral circulatory
complications)

[DA7Z](#) *Diseases of the stomach or the duodenum,
unspecified*

(other specified
complications)

[1A40](#) *Gastroenteritis or colitis without specification of
infectious agent*

[1G40-1G41](#) *Sepsis*

[1C41](#) *Bacterial infection of unspecified site*

[1F28](#) *Dermatophytosis*

[1F2D](#) *Non-dermatophyte superficial dermatomycoses*

[1F23](#) *Candidosis*

[3B20](#) *Disseminated intravascular coagulation*

[5A41](#) *Hypoglycaemia without associated diabetes*

[5C80.00](#) *Primary hypercholesterolaemia*

[5C80.1](#) *Hypertriglyceridaemia*

[5C80.2](#) *Mixed hyperlipidaemia*

[5C80.Z](#) *Hyperlipoproteinaemia, unspecified*

[5C76](#) *Hyperkalaemia*

[5D2Z](#) *Metabolic disorders, unspecified*

[8A42.Y](#) *Other specified acute disseminated
encephalomyelitis*

[8A42.Z](#) *Acute disseminated encephalomyelitis, unspecified*

[BA00.Z](#) *Essential hypertension, unspecified*

[BA01](#) *Hypertensive heart disease*

[BA40-BA6Z](#) *Ischaemic heart diseases*

[BB40-BB4Z](#) *Acute or subacute endocarditis*

[BC0Z](#) *Heart valve diseases, unspecified*

[BC43.0Z](#) *Dilated cardiomyopathy, unspecified*

[BC43.Z](#) *Cardiomyopathy, unspecified*

[BC81.3](#) *Atrial fibrillation*

[BC81.20](#) *Cavotricuspid isthmus dependent macroreentry
tachycardia*

[BC81.2Z](#) *Macro reentrant atrial tachycardia, unspecified*

[BC60](#) *Atrial premature depolarization*

**TUC diabetic complication
in ICD-11**

If desired, postcoordination may be used to specify complication of diabetes mellitus

[BC61](#) *Junctional premature depolarization*

[BC70](#) *Ventricular premature depolarization*

[BC71.1](#) *Ventricular fibrillation*

[BC80.20](#) *Sick sinus syndrome*

[BC9Y](#) *Other specified cardiac arrhythmia*

[BC9Z](#) *Cardiac arrhythmia, unspecified*

[BD10-BD1Z](#) *Heart failure*

[BE2Z](#) *Diseases of the circulatory system, unspecified*

[8B00](#) *Intracerebral haemorrhage*

[8B02](#) *Nontraumatic subdural haemorrhage*

[8B03](#) *Nontraumatic epidural haemorrhage*

[8B0Z](#) *Intracranial haemorrhage, unspecified*

[8B11](#) *Cerebral ischaemic stroke*

[8B20](#) *Stroke not known if ischaemic or haemorrhagic*

[8B22.Y](#) *Other specified cerebrovascular disease*

[8B2Z](#) *Cerebrovascular diseases, unspecified*

[8B25.1](#) *Late effects of intracerebral haemorrhage*

[8B25.3](#) *Late effects of other nontraumatic intracranial haemorrhage*

[8B25.0](#) *Late effects of cerebral ischemic stroke*

[8B25.4](#) *Late effects of stroke not known if ischaemic or haemorrhagic*

[8B25](#) *Late effects of cerebrovascular disease*

[8D40.1](#) *Neuropathy due to nutritional deficiency*

[8D40.2](#) *Myopathy due to nutritional deficiency*

[8D40.Y](#) *Other specified neurological disorders due to nutrient deficiency*

[8D40.Z](#) *Neurological disorders due to nutrient deficiency, unspecified*

[BD30.20](#) *Acute thromboembolic lower limb arterial occlusion*

[BD30.0](#) *Acute upper limb arterial occlusion*

[BD30.2](#) *Acute lower limb arterial occlusion*

[BD70](#) *Superficial thrombophlebitis*

[BD72](#) *Venous thromboembolism*

[CA40.1](#) *Viral pneumonia*

[CA40](#) *Pneumonia*

TUC diabetic complication in ICD-11

If desired, postcoordination may be used to specify complication of diabetes mellitus

[DA60-DA63.Z](#) *Ulcer of stomach or duodenum*

[ME24.Y](#) *Other specified clinical manifestations of the digestive system*

[1B70.3](#) *Ascending bacterial lymphangitis*

[1B70.Y](#) *Bacterial cellulitis or lymphangitis due to other specified bacterium*

[1B70.Z](#) *Bacterial cellulitis or lymphangitis due to unspecified bacterium*

[EB21](#) *Pyoderma gangrenosum*

[EA3Z](#) *Unspecified skin disorder attributable to viral infection*

[EA89](#) *Generalised eczematous dermatitis of unspecified type*

[EH90](#) *Pressure ulceration*

[FA26.2](#) *Chondrocalcinosis*

[1B71](#) *Necrotising fasciitis*

[GC08.Z](#) *Urinary tract infection, site and agent not specified*

3.7 Innovation to morbidity coding in ICD-11

‘Present on Admission’

The inclusion of the new Extension codes in ICD-11 provides capacity for coding qualifying information of linked stem codes. Among the new qualifying features is the particularly important status display feature that allows for distinction of diagnoses present at admission from diagnoses arising after hospital stay began.

3.8 Functioning section

An optional functioning section has been embedded in ICD-11 to enable the classification and measurement of the impact of health conditions in terms of functioning. For further information see section [1.1.6.2](#).

Overall, the linkage between ICD-11 and ICF may assist with the following use cases:

- evaluation for general medical practice (e.g. work in capacity assessment)
- evaluation for social benefits (e.g. disability pension)
- payment or reimbursement purposes
- needs assessments (e.g. for rehabilitation, occupational assistance, long term care)
- outcome evaluation of interventions

Wherever full *functioning* reporting is desired and required, the ICF should be used.

3.9 General features of ICD–11

The main structural innovation of ICD–11 is that it is built on a Foundation Component from which the tabular list can be derived. The international reference tabular list is the statistical classification for morbidity and mortality. Due to the addition of a Foundation Component, and the electronic design of ICD-11, a new terminology had to be introduced that had not been used in prior versions of ICD. (See Section [2.13](#) for further information.)

For more detail on the terminology of ICD-11, see Section [1.1.6.1 Integrated use with Terminologies](#).

3.10 Traditional Medicine conditions - Module 1 (TM1)

A large percentage of the world's population uses some form of Traditional Medicine. However, standardised data and information on health status of these users remain largely absent from national and international health data collections. The use of Complementary and Alternative Medicine (CAM) therapies has become a huge industry and is expected to grow. As a result of this gap in information about TM and the size of the industry, resources have been invested in the creation of a classification tool to allow data to be collected and analysed.

ICD-11's supplementary chapter on Traditional Medicine disorders and patterns is designed to be integrated with coding of cases in conjunction with the Western Medicine concepts of ICD Chapters 1-25.

As with other ICD chapters, the TM1 chapter is a tool for classifying, diagnosing, counting, communicating and comparing TM conditions, it will also assist research and evaluation to assess the safety and efficacy of TM. This chapter not judging TM practice or the efficacy of any TM intervention.

This chapter must not be used in mortality coding.

3.11 Preparations for the Eleventh Revision

By 2003, it was becoming clear to the WHO and the Collaborating Centres that a further revision of the ICD could not be long delayed. The extent to which ICD updating could encapsulate emerging developments was limited by the structure of ICD–10, and some issues needed extended development and discussion with expert groups. A special meeting of Collaborating Centres in Helsinki in 2004 discussed the need for a revision and issues to be addressed as part of the revision process. The 2004 WHO-FIC meeting subsequently adopted a revision process work-plan which was progressively developed at ensuing meetings.

In 2007, the WHO formally launched the revision process. Oversight has been provided through a broad-based Revision Steering Group. Technical work has been undertaken by a series of Technical Advisory Groups, with cross-cutting groups examining mortality, morbidity and quality and safety issues. For the first time, a chapter on description of diseases and patterns of diseases from a Traditional medicine standpoint has been included.

A Content Model, including a range of components for each ICD entity has been developed, giving a rich foundation for the ICD. Other classifications and terminologies are linked or

included where possible to ensure ICD is aligned with them, and items used in other members of the WHO Family of Classifications have been aligned wherever possible. The more traditional statistical classification for mortality and morbidity is obtained from the Foundation Component of ICD-11 as a Tabular list. Extension codes are used to limit content volume but still allow detailed classification of disease entities. Supplementary chapters and sections allow capturing on an optional basis information about traditional medicine diagnoses and functioning. Based on the experiences with ICD-9 and ICD-10 an updating mechanism was designed, that allows improvements in user guidance and scientific updates without compromising the statistical use of the classification.

The revision of ICD-11 has taken place in several phases. First, a list of issues that were known from the use of ICD-10 and that could not be solved in its classification structure was compiled and possible solutions were formulated.

Second, input was received from many scientific groups in the key subject areas, with a focus on clinical perspectives.

Finally, centralised editing occurred, aimed to adjust imbalances in content generated by multiple, independently operating expert groups in the previous phase of the revision, and to ensure the overall structure is consistent and practicable for users for mortality and morbidity statistics. The final version also received input from field testing, Member State comments, and ongoing submission and processing of proposals.

3.12 Annex A: ICD-11 Updating and Maintenance

This Annex describes the review process, the release cycles and the proposals workflow for updating ICD-11.

Official releases of the ICD-11 classification are produced annually for international use in mortality and morbidity (this is known as the ‘blue browser’). By contrast, the ICD-11 Foundation Component is continuously updated. The updating is carried out at different levels with corresponding different frequencies.

The ICD-11 is being released in five-yearly ‘stable’ versions for international use (contains updates that impact on the four- and five- character structure), unless urgent public health needs require otherwise. The releases are supplemented with version identifiers that are used for reporting in conjunction with the codes. Transition tables and materials showing the differences are provided with every version.

Updates at a more detailed level than four- and five- characters can be published annually. Small error corrections that serve to clarify meaning, indexing or errors, may be communicated annually. Additions to the index can be done on an ongoing basis.

Mortality and morbidity rules will be updated in longer term cycles of every 10 years.

All countries that have implemented the ICD-11 are encouraged to adopt the updates in order to ensure greatest possible standardisation of coding results. If a country for whatever reason cannot implement a certain year of updates it shall at least ensure that the reported data is in line with the most recent version of ICD-11.

The WHO has taken all reasonable precautions to verify the information contained in the ICD and its different versions and editions. However, the ICD is being distributed without warranty of any kind, either expressed or implied. The responsibility for the interpretation and use of ICD lies with the user. In no event shall the WHO be liable for damages arising

from its use.

The publisher of ICD-coded information is liable to ensure proper use of the ICD and present clearly the methodology for data collection and mechanisms that were used to modify the original data in order to indicate the comparability of the presented outcomes. For mortality data, no deviation from the methodology indicated in the ICD is permitted.

For information on taxonomy and structuring see [1.2](#).

3.12.1 Background

Updating ICD-11 makes sure that ICD meets the needs of users in content and terminology.

Updating and decisions follow the principles laid out in [1.2.1 Taxonomy](#).

Updates include changes to chapters 01-24 of ICD, the supplementary chapter for traditional medicine conditions (including the development of additional modules), the supplementary section for functioning assessment, the extension codes, special views and special tabulation lists, elements of the foundation of ICD, as well as changes to the reference guide.

All proposals are entered on an online maintenance platform, for verification of completeness, discussion and editing. The platform provides the infrastructure for routing proposals to reviewers and experts, and for providing feedback to the original authors. The maintenance platform also shows the final outcome of the proposal that has been entered in the authoring platform and become part of the ICD. Any individual user of the classification can submit a proposal for an update to the ICD. Such updates can refer to one or more entities of the ICD. They may address the position of entities in a tabular list, in the Foundation Component, and any element of the content model. The maintenance platform of ICD-11 (known as the 'orange browser') is used for proposals and comments. Any input to ICD-11 and its components requires proper referencing of sources, details of scientific evidence, and permission from the owner of any copyright materials (where applicable).

The proposals will be reviewed by scientific experts and classification experts. The decision regarding the outcome of a particular proposal will be based on the recommendations by these experts.

A workflow between a mortality reference group (MRG) and a morbidity reference group (MbRG), a medical scientific advisory committee (MSAC), and the overall overseeing classification and statistics advisory committee (CSAC) will ensure that all aspects concerning a proposal are considered. Reviews of the synthesis by classification experts ensure suitability of the proposed changes to the diverse use cases of the ICD.

The process is based on consensus of the members of the CSAC about a proposed change. All rounds of editing will be handled through electronic platforms. Where consensus cannot be achieved, the proposal can either be deferred to subsequent cycles of editing pending arbitration by the WHO or be solved in a face-to-face meeting of classification and content experts. In all other cases, a consensus recommendation is given to WHO for final decision.

All proposals for change must be submitted through the proposal mechanism to ensure a clear and transparent review of the proposed content. The different types of proposals that may move through a workflow in order to ensure consistency, structural integrity, and

scientific correctness of the classification. The different workflows warrant proper use of the available resources of the WHO Family of International Classifications WHO-FIC Network and WHO. All changes are reported. Several steps may be needed for verification of updates.

3.12.2 Updating Cycle

The updating is carried out at different levels with different frequencies. That will keep stability for mortality and allow quicker updates for morbidity use.

- Updates that impact on international reporting (the four and five-character structure of the stem codes) will be published every five years.
- Updates at a more detailed level can be published at annual rates and pending the needs of clinical modifications also twice a year.
- Additions to the index can be done on an ongoing basis.
- Mortality and morbidity rules will be updated in a 10-year cycle.
- Other updates to the reference guide can be published at annual rates.

3.12.2.1 Types of proposals for ICD-11-MMS maintenance

After reviewing the established proposal types and criteria of ICD-10 and those used thus far during ICD-11 Revision, taking into context the needs of ICD-11, the following proposal types for ICD-11 are proposed (see Table below for impact on Morbidity and Mortality Statistics (MMS)).

- **Add new entity:**
 - to add an entity 'below the shoreline' (becoming an index entry in MMS)
 - to add an entity 'above the shoreline' (becoming a category in MMS)
- **Delete entity:**
 - applicable to an entity below the shoreline (removing an index term from the MMS)
 - applicable to an entity above the shoreline (removing a category from the MMS)
- **Change of Coding Status:**
 - moving an entity from the index into the MMS (e.g. giving it an individual code), or an entity from the MMS to the index (e.g. eliminating the individual code and directing it elsewhere)
- **Content Enhancement** including the following subtypes:
 - Change of Preferred Term (title) (Changes to a code that affect the meaning of that code are not allowed and will not be accepted. If a concept is

outdated and must be updated, or a new concept is necessary, the relevant delete entity or add new entity proposal types, or both, must be used.)

- Addition / Deletion of a synonym
- Addition / Deletion of an exclusion
- Change of Description (Definition)
- Change of Additional Information (Long Definition)
- Correction of spelling or grammar (in any field)
- Addition / Deletion of a mandatory postcoordination combination of stem codes
- Addition / Deletion of entity rubric content (does not allow change to meaning)
- **Structural Change** including the following subtypes:
 - Change a primary parent link
 - Change a secondary parent link
- **Reference Guide Change** applicable to any text of the ICD-11 Reference Guide, including coding rules and defaults, etc. Subtypes include:
 - Correction of spelling or grammar
 - Clarification of a rule
 - Change to a rule (that effects data integrity), including changing a coding hint
- Proposals for clarification that do not require change to the classification
- Correction of inconsistency between volumes

Ideally, each proposal will specify if it relates to the foundation or to a classification, including if it is for a national clinical modification or specialty linearisation. Tick boxes in the proposals will allow to indicate the scope.

Not all authors will be familiar with these distinctions. The default may assume that the proposal relates to the ICD-11 MMS, unless specified otherwise. Regardless, the CSAC will need to determine if the item, once accepted, will be ‘above or below the shoreline’ in the ICD-11 MMS.

Not all proposals will require the same level of scrutiny, as each may be considered in the context of its impact on the statistical classification and the desirability for a scientifically accurate and up-to-date classification. A ‘fast track’ to review urgently needed updates, such as for national clinical modifications will be put in place as needed. Criteria and a specific workflow exist for each ‘track’ used for proposals.

Table 1: Overview of ICD-11 maintenance proposal types and their potential impact on MMS-collected data. List does *not* imply a hierarchy of prioritization. ‘X’ means applies.

Proposal Type	Major	Minor	None
Add new entity	X	X	
Delete entity (+)	X		
Change of Coding Status	X	X	
Content Enhancement			
Change of Preferred Term (title)	X	X	X
Addition/Deletion of a synonym		X	
Addition/Deletion of an exclusion		X	
Change of Description	X	X	X
Change of Additional Information (Long Description - outside WHO)			X
Correction of a typo (in any field)			X
Addition/Deletion of mandatory postcoordination combination of stem codes	X		
Addition/Deletion of entity rubric content – no change to meaning			X
Structural Change			
Change a primary parent	X		
Change a secondary parent			X
Reference Guide change			
Correction of a typo			X
Clarification of a rule			X
Change to a rule (that affects data integrity), including changing a coding hint	X		

(+) Errors are deleted; obsolete terms and entities are not deleted as they assist with mapping and coding as they may still be in use

Table 2: Groups responsible for maintenance of potential changes.

Proposal Type	CSAC	MSAC	MRG	MbRG	FDGRG or TM
Add new entity	X	X	X	X	X+
Delete entity (+)	X	X	X	X	X+
Change of Coding Status	X	X	X	X	
Content Enhancement					
Change of Preferred Term (title)	X	(X)			X+
Addition/Deletion of a synonym**					X+
Addition/Deletion of an exclusion**					
Change of Description	X	X	X	X	X+
Change of Additional Information (Long Description - outside WHO)		(X)			X+
Correction of a typo (in any field)**					
Addition/Deletion of mandatory postcoordination combination of stem codes	X		X	X	
Structural Change					
Change a primary parent	X	(X)	X	X	X+
Change a secondary parent	X	X			
Reference Guide change					
Correction of a typo**					
Clarification of a rule	X		X	X	X+
Change to a rule (that affects data integrity), including changing a coding hint	X	(X)	X+	X+	X+

(+) errors are deleted; obsolete terms and entities are not deleted as they assist with mapping and coding as they may still be in use. (X) applies only in special situations. X+ applies only for the specific use case. ** WHO team

3.12.3 Proposal completeness

Any individual can submit a proposal for an update to the ICD. Proposals shall be provided in the format of a short (approximately 500-words) explanation with references to underpinning literature and evidence (publications in peer reviewed journals, or in official meetings of WHO, its CC or NGO in official relationships). The proposal shall also visualize the changes in the position and address potential impact on entities outside the proposal.

- The author has to register with full name and affiliation and declare any possible conflict of interest.
- All proposals must have a clearly written and compelling rationale, with citations to establish the proposals' evidence base.
- Proposals that suggest adding entities must have a description, and a description of the entity. This ensures the correct placement in the foundation. The rationale must

have a scientific background, with references to publications in peer reviewed journals, or in official meetings of WHO, its CC or NGO in official relationships.

- Proposals for new codes should include information about how the case would be coded if the proposed new code is not accepted.
- Proposals with impact on the statistics must include a description or analysis of the resulting impact like frequency.
- Proposals suggesting rule changes must come with an impact analysis.
- An incomplete proposal will be returned to the author.
- The proposal mechanism will not allow submitting proposals without rationale or with missing description or definition, adequate to process the proposal.

3.12.4 Proposal Timelines

Proposals can be submitted at any time. No impact proposals are processed on an ongoing basis, proposals requiring review by any of the groups and committees involved in the workflow, are bundled every 28 February of a year and routed in the necessary workflow.

Proposals are processed in parallel by the relevant groups. Formal comments are provided in two rounds (two months, one month) -offering the opportunity for edits in between. Final decision about acceptance, rejection or 'further discussion' is taken at a teleconference of the CSAC at the WHO-FIC annual meeting in October every year. Official releases are published end of September for validity according to the updating cycle of the kind of proposal, earliest being proposals for adoption in January of the following year (minimum six months for translation, three months for formal dissemination, e.g. for clinical detail, secondary parents or synonyms).

Review and Release Schedule



Proposal and Release Timelines